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REVIEW ARTICLE

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The use of electronic tongue and sensory panel on taste evaluation of pediatric medicines: a systematic review

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ABSTRACT

The palatability of medications is an essential factor for children's adherence to drug treatment. Several methods for drug taste assessment have been developed. The aim of this review is to explore the literature reports of the main methods for the evaluation of medicines taste, named electronic tongue (e-tongue, *in vitro*) and human sensory panel. A systematic search was performed up to March 2020 and a total of 88 articles were selected. The e-tongue (57.5%) has been more frequently described than the sensory panel (10.3%), while some articles (32.2%) used both techniques. 74.7% of the articles mentioned 'pediatric', 'paediatric' or 'children' in the text, but only 19.5% developed formulations targeting pediatric audience and sensory testing in children is rarely seen. The e-tongue has predominance of use in the taste evaluation of pediatric medicines probably since it is fast, easy to perform and risk free, besides presenting less imprecise data and no fatigue. The human panel is more realistic, despite its intrinsic variability. In this sense, it is proposed the use of e-tongue as a fast way to select the most promising sample(s) and, after that, the sensory panel should be applied in order to confirm the taste masking.

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Electronic tongue; pediatric dosage forms; sensory panel; systematic search; taste evaluation

Introduction

Medication palatability is defined as the general acceptance of a drug in terms of its smell, taste, texture, dosage, volume and flavor (Ternik et al. 2018). It has a fundamental role in the development of oral pharmaceutical formulations, presenting a direct relation to the acceptability and, consequently, to patient adherence to the treatment. Organoleptic disadvantages of medications are often neglected by researchers and the pharmaceutical industry, regardless of the impact of these issues on patient compliance (Aksöz et al. 2019). The development of palatable dosage forms, especially for chronic diseases in children, can be considered a major challenge (Zajicek et al. 2013).

Palatability is recognized as one of the most decisive factors influencing adherence to therapeutic protocols in pediatric patients (Thompson et al. 2015; Bastiaans et al. 2017). Palatability influences directly the willingness of children and their ability to ingest medicines (Bastiaans et al. 2017; Rao et al. 2019). Therefore, the development of palatable formulations for children becomes necessary, it being required to measure the taste masking capacities. The choices of the methods used for sensory evaluation varies with the age of the children (Bastiaans et al. 2017; Lopalco et al. 2020). It must be considered that the taste preferences of adults and children are different, as well as between healthy and sick children (Bastiaans et al. 2017).

Quantitative assessment of palatability is an important component in the formulation development process (Anand et al. 2007; Immohr et al. 2017a). Human sensory testing is one of the main methods used to evaluate the palatability of oral formulations (Yıldız et al. 2018; Rao et al. 2019). However, the use of human volunteers for testing requires volunteers to be trained and the research to be conducted in accordance with internationally accepted ethical parameters (Immohr et al. 2017a; Fan et al. 2020). The taste of pediatric medications is rarely studied in children due to restricted regulation, toxicity and safety reasons (Anand et al. 2008). Currently, electronic tongue (e-tongue) is described in the literature as the main method for evaluating the palatability of medications (Breitkreutz and Boos 2007; Haraguchi et al. 2016; Zidan 2017; Chikukwa et al. 2020).

This investigation presents a systematic literature review on the use of e-tongue and human sensory panel to evaluate the taste of medications, especially those related to pediatrics. The differential of the present review is the covering of both methods: sensory panel and e-tongue, as complementary tools to evaluate taste masking of drugs for infant patients.

This review aims to answer the following questions: (a) Which studies are being published regarding the evaluation of medication taste by e-tongue and human panel; and among them, which ones include specifically the pediatric public? (b) How are these studies performed? (c) Which pharmaceutical dosage forms and drugs are more cited in these studies? (d) Electronic tongue or sensory panel, which is the best method for taste evaluation of pediatric medicines? Is it possible to replace one with another?

Methods

Eligibility criteria

The inclusion criteria for article selection were: (1) papers that used an e-tongue method and/or human sensory panel

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evaluation for the evaluation of medication taste; (2) the full text was available; and (3) it was written in English. The exclusion criteria were: (1) articles that evaluated the flavor of food; and (2) review articles.

Search strategy

Research articles that used e-tongue and/or sensory analysis in the assessment of medication taste were searched in the Web of Science and PubMed database using the keywords combination: taste* AND mask* AND drug* AND sensor* AND panel* to identify records for sensory panel; or taste* AND mask* AND drug* AND electronic* AND tongue* to identify records for electronic tongue. The research was conducted without period of restriction until March 2020.

Study selection process and data extraction

The process used to select articles was the screening of titles and abstracts, based on the inclusion criteria, with a check from a second reviewer (following PRISMA guideline). The Figure 1 provide a flow diagram outlining the methodology applied in our systematic review.

After the initial search, papers that included an e-tongue evaluation and/or a sensory panel for the evaluation of medication taste were selected. From the articles chosen, the terms 'pediatric', 'paediatric', 'children' and 'child' were identified when present.

Results and discussion

Taste evaluation techniques

The taste of a drug is a main contributor to patient satisfaction and treatment compliance, especially in children (Mistry and Batchelor 2017). Many drugs have a bitter taste, which represents a challenge for the development of oral dosage forms (Mennella et al. 2013). There are several flavor evaluation techniques, both *in vitro* and *in vivo* methods that can be used to evaluate active substances or formulations (Yajima et al. 2002). Among the most used techniques are the *in vivo* evaluations using human volunteers and *in vitro* studies using sensors known as e-tongue.

Figure 2 shows the number of articles published based on the keywords searched and according to the year. It is possible to detect an increase of publications in the last few years. 88 articles were chosen after the refinement and, among these, 50 (57.5%) used the e-tongue, 9 (10.3%) used the sensory panel and 28 (32.2%) used both techniques to evaluate the taste of medications. These results indicate the greater use of the e-tongue in detriment of the sensory panel. It should be highlighted that the percentage of articles that used both techniques to confirm flavor evaluation was higher than the number of articles that used only the sensory panel.

Sensory panel analysis

Table 1 shows the articles that used sensory panel. According to Sawan (2015), the human panel test is a psychophysical assessment of gustatory stimuli. This method is conducted in a controlled manner, where tasters (foods, chemicals, drugs) are evaluated estimating gustatory responses, usually in healthy, adult, human volunteers (Woertz et al. 2011a; Lawless 2012).

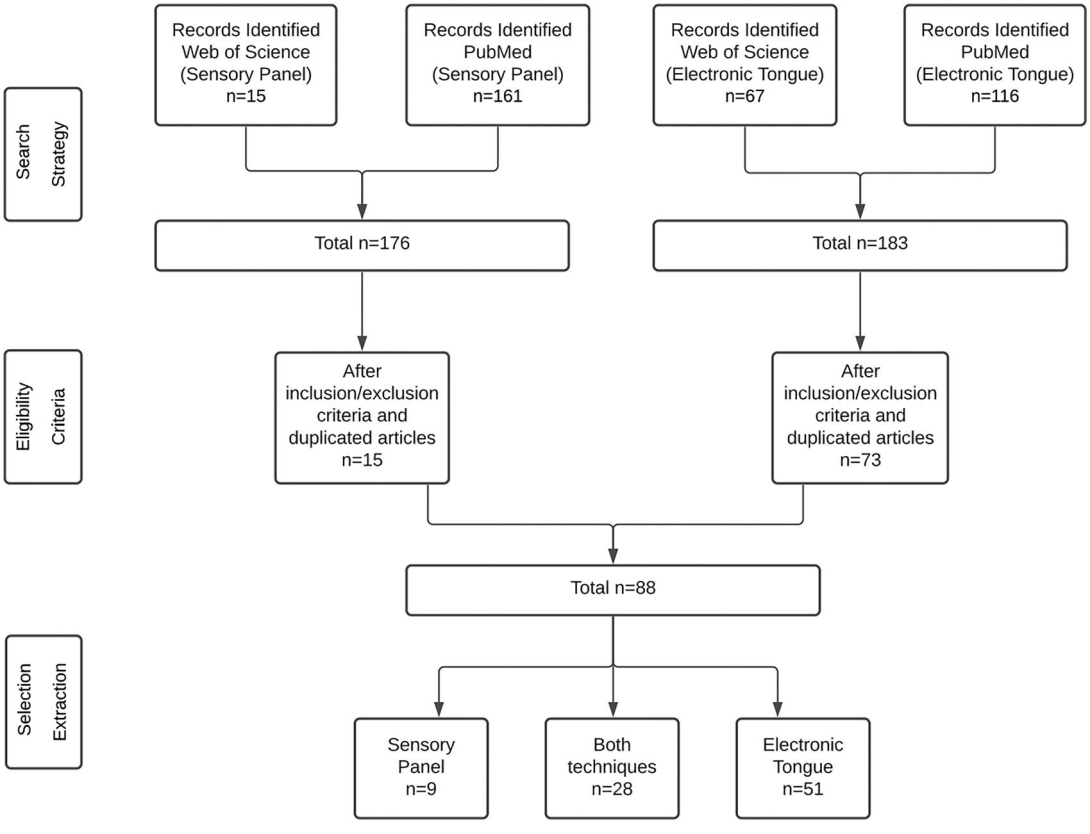


Figure 1. Flow diagram for the systematic review.

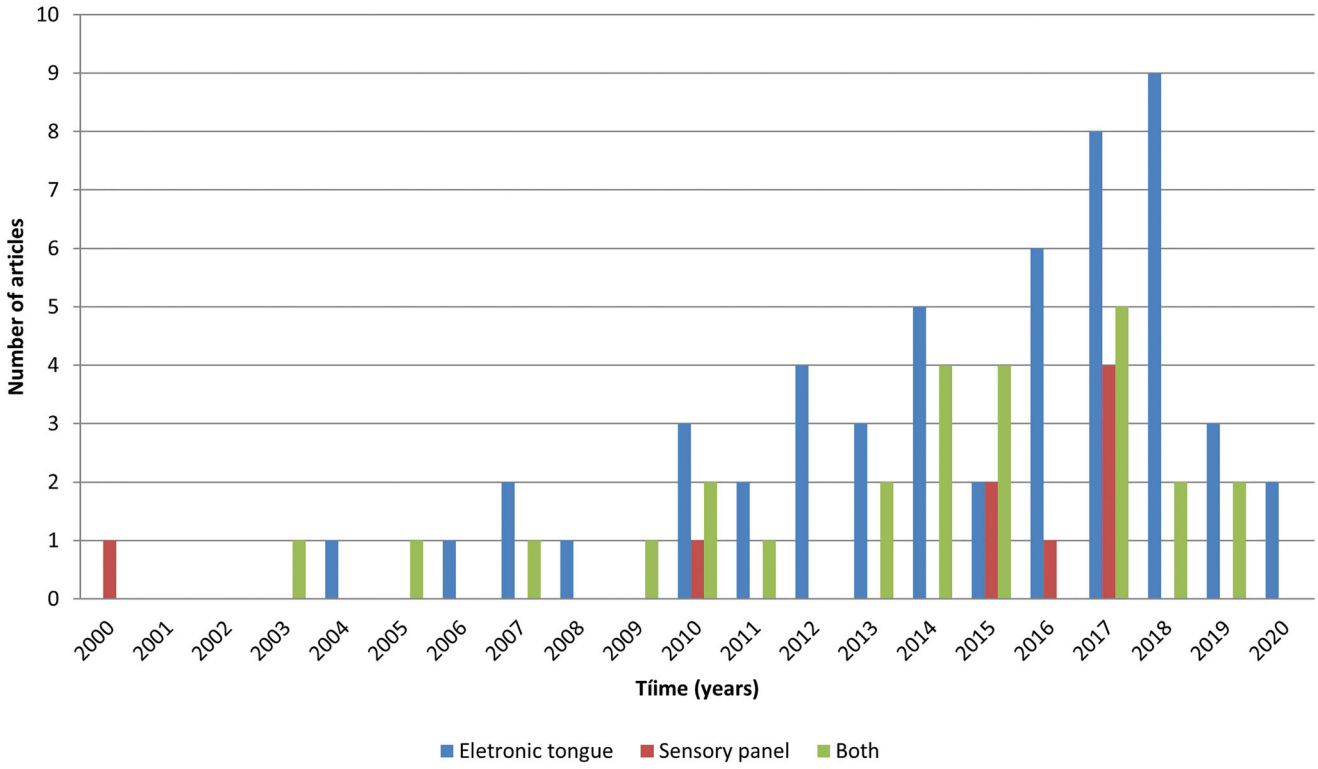


Figure 2. The profile of publications regarding taste evaluation of medications by human panel and/or e-tongue over the years. The year 2020 was analyzed until March.

Taste is a very subjective perception. Depending on the individuals, the perceived taste may vary at different levels. However. if the experimental procedure is well controlled, it is possible to measure taste thresholds accurately and reproductively. Several flavor evaluation scales are employed for the evaluation of formulations, for example, hedonic scales, visual analog scales or the combination of both (Mistry and Batchelor 2017).

In this perspective, the evaluation methods used to measure human responses to sensory stimuli have three main objectives: (1) to identify the presence of perceptible differences between samples, (2) to quantify important sensory characteristics for the product, and (3) to verify particular problems that cannot be detected by other procedures. They can be classified into two categories: analytical methods (divided into discriminative and descriptive methods) and affective methods (also called preference methods) (Anand et al. 2007).

Discriminatory methods are used when it is desired to determine whether there are significant sensory differences between products with different characteristics. These methods are also used when determining thresholds of perception of a stimulus. On the other hand, the descriptive methods have the objective of describing the sensorial attributes of products in a qualitative form (through verbal terms that describe the sensorial properties of the evaluated product), or in a quantitative assessment (to measure the perceived intensity of these attributes, by means of scales, for example) (Mennella et al. 2015).

Affective methods aim to answer the degree of appreciation of consumers and which product is preferred among the options (Mennella et al. 2015). They are also called Consumer Testing, when performed with a larger number of judges (over 50), selected to represent the target population. The main objective of affective tests is to evaluate the personal response of current or potential consumers in relation to a prototype or product. In affective analysis, the selection of the participant team is not related to sensory abilities, such as discriminating or describing capabilities, but rather, the test usually includes the target population for which the product is intended (Anand et al. 2008; Mennella et al. 2015).

Affective methods, therefore, fall into two categories: preference tests and acceptance tests. The preference tests are used when it is desired to identify the preferred or least preferred of two or several products, either for development improvement or to compare similar competing products. The preference test requires the analyzer to choose one item over another. The questions asked are, for example: 'Which sample do you prefer?' or 'Which sample do you like the most?' (Mennella et al. 2015).

The entire physiological process of successful taste masking analysis occurs through the interaction of the stimulus with receptors incorporated into the membrane of the taste buds receptor-stimulus; interactions are sent as an electrical signal through the appropriate nerve fibers to the brain, thus, taste is detected and sensed (Yajima et al. 2002; Woertz et al. 2011b).

Responses are sensitive taste measurements and are designed statistically to minimize bias and varying responses among human volunteers. In a typical human panel, a group of about 5–10 human volunteers is trained for flavor evaluation using reference solutions ranging from tasteless to very bitter. Numerical values are then assigned to these bitterness levels (e.g. 0–5) and are statistically evaluated to detect if there is a significant difference in the results obtained. Subsequently, the test solution is tested and evaluated on the same scale to assess its bitterness (Mennella et al. 2015).

Volunteers must be trained in the basics of sensory identification. This step is necessary to understand the correct way to conduct the evaluation of the taste, disregarding their personal preferences or any influence that may be considered a bias (Taj et al. 2014). Training should include application of the sample, tasting, intensity rating and use of an appropriate scale

Table 1. Articles that used sensory analysis through a human panel to evaluate the taste masking of medications according to searched keywords.

Drug(s)	Dosage form(s)	Objective	Human panel	Scale	Reference
Caffeine, propylthiouracil, quinine and urea	Solution	To verify if the bitter taste of some substances would be reduced with the addition of sucrose	Women (109) and their children (81 girls and 60 boys), all of whom were between the ages of 5 and 12years	The choice was made by forced paired method comparison. To better understand individual differences, the adults also measured each solution using a labeled magnitude scale	(Mennella et al. 2015)
Capsaicin	Solution	The sensory analysis was performed with trained students to evaluate the pungency after nanoencapsulation	Sensory panel composed of 24 students (13 males, 11 females), with ages ranging between 19 and 27 years	The scale ranged from 0 to 10 (0 corresponds to 'not detectable' and 10 to 'very intensive'.	(Kaiser et al. 2015)
Griseofulvin and caffeine anhydrous	Solution	To investigate the potential of Kleptose Linecaps DE17 in masking the bitter taste of drugs by hot melt extrusion	9 volunteers (5 men and 4 females in the age group 18–42) were selected for testing caffeine formulations, and 6 volunteers (3 male and 3 females in the age group 18–47) for testing griseofulvin formulations	The subjects were requested to score the bitterness on a scale of 0–5 for each product where 0 indicates none and 5 indicates strong bitterness	(Juluri et al. 2016)
Ofloxacin	Solution	To evaluate the flavor masking of ofloxacin solution by means of ion exchange resins	Sensory panel taste assessment was performed by tests on 9 participants (ages 18–30)	The bitterness was recorded on the bitterness scale ranging from 0 to 5 (0-no bitter, 1-threshold bitter, 2-slight bitter, 3-moderate bitter, 4-bitter and 5-strong bitter)	(Rajesh and Popat 2017)
Prednisone	Oral liquid (not specified)	To use a new drug release technology to create a masked taste prednisone formulation	Sensory panel taste assessment was performed by testing on 24 participants (age 18–40)	Validated Hedonic Scale, where the rankings are converted to numerical values	(Bai et al. 2017)
*Lopinavir/ritonavir	Liquid (not specified)	To document the range of individual differences in taste and palatability of a liquid formulation	Adult sensory panelists (84 women). Panelists averaged 36.4 (0.9) years of age (range, 23–58 years of age)	General Labeled Magnitude Scale and Hedonic scale for across group comparisons.	(Mennella et al. 2017)
Acetaminophen	Liquid (not specified)	To develop a numerical model from sensorial response curves obtained by a panel and the application of a compartmental model to study the sensorial response	3 trained individuals (ages not declared)	Bitterness intensity values were based on a 0–3 scale, with 3 being strong bitterness, 2 being moderate, 1 being slight	(Saavedra and Cuadra 2000)
Chlorpheniramine maleate	Orally disintegrating film	To develop films using hot extrusion technique and to incorporate sweeteners to improve the palatability	Sensory panel taste assessment was performed by testing on 6 humans of both sexes, with ages between 18 and 42 years	The range of perception varied from 0–5: 0-no taste, 1-flavor of something, 2-slightly bitter, 3-moderately bitter, 4-bitter and 5-strongly bitter	(Pimparade et al. 2017)
Ornidazole	Microspheres compressed tablet	To achieve the taste masking of ornidazole by preparing rapidly disintegrating tablets from microspheres using Eudragit® E-100	Sensory panel taste assessment was performed by testing on 20 healthy human volunteers (20–30 years)	The range of perception varied from 0–5 (0-good, 1-tasteless, 2-slightly bitter, 3-bitter, 4-very bitter, 5-awful)	(Shishu and Kapoor 2010)

(Thompson et al. 2015). Panel members are tested for their ability to taste and express sensory perceptions. The performance of the members of the sensory analysis panel is regularly validated (Anand et al. 2008). Human sensory panels can also use untrained volunteers. However, naive volunteers tend not to detect slight differences between samples (Giaretta et al. 2019).

In the study of Mennella et al. (2017), the types of bitterness and irritation of the antiretrovirals analyzed in this study was different among the participants. Those who experienced less bitter and sweet taste sensations had a genetic signature different from the other participants. With this study, it can be concluded that the perception of bitterness changes greatly from person to person, according to its genetic signature.

Regarding the sensory panel, most of the studies found during literature search only mention the importance of making medicines aimed towards children and do not use the sensory panel with children, but rather with young adults. This is probably due to ethical issues and limited understanding of children. Mennella et al. (2015) carried out their investigation with children from 5 to 12 years old and with adults. Of the 141 children, 22 (15.6%) did not understand the task or could not comply with the study procedures and therefore were excluded from the final analysis. The younger children, around 5 years old, were the most excluded.

In the investigation by Pimparade et al. (2017), the human panel study confirmed the reduction of bitterness of chlorpheniramine maleate (antihistamine) films compared to the pure drug and to the physical mixture of the drug. In this study, the authors did not use sensory panel with the children, only cited 'pediatric' in the keywords, introduction, and conclusion. Another example is the study of Kaiser et al. (2015) which had a panel of 24 students, aged between 19 and 27 year sold, who analyzed the pungency of nanoencapsulated capsaicin. Shishu and Kapoor (2010) have also been successful in their study using adults to analyze fast disintegrating ornidazole tablets (antibiotic), which are disintegrated in the oral cavity in about 30 s, presenting a pleasant taste.

The results obtained by Bai et al. (2017) indicated that a prednisone formulation was indistinguishable from the control, and both were preferable to the reference formulation in every category of palatability assessed using the Hedonic Scale. This study cites 'pediatric' in the introduction, discussion, and conclusion. Herein, children are also not included in the panel (24 healthy young adults).

The human panel evaluation is a standard method for measuring the extent of taste masking and approval of medications. However, factors such as personal taste sense differences, drug toxicity and individual taste discernment, among others, can have an impact on taste assessment (Jain et al. 2010). Besides the reliability of human panel tests, there is a limitation raised by ethical and safety concerns, especially when the drug tested has limited toxicity data (Duong et al. 2002).

Electronic taste analysis

'Electronic tongue is a flavor sensor that simulates human perception through a matrix of chemical sensors and statistical data processing' (Vlasov et al. 2005). The system is composed of probe's membranes that captures the flavor and a signal transducer that provides the results (Raman et al. 2011).

The most commonly used e-tongue in the pharmaceutical industry is composed of multi-channel flavor systems (Eckert et al. 2014). There are several electronic sensors which are used to evaluate taste, using different measurement principles, such as voltammetry, potentiometry, impedance and optical mass-emitting techniques (Duong et al. 2002). E-tongues are systems that use an array of sensors capable of detecting individual substances as well as complex mixtures through membrane sensors and electrochemical techniques. Several challenges, such as possible toxicity and subjectivity of the flavor evaluators in the sensory panel, can be overcome by e-tongue (European Medicines Agency 2006; Rachid et al. 2010; Petrovick et al. 2018).

From the 88 articles chosen, 65 articles (73.9%) mentioned the terms 'pediatric', 'paediatric' and/or 'child' in some part of the article, whether in the title, abstract, introduction, keywords and/or conclusion, showing the importance of developing a formulation appropriate for this target public. However, only 11.1% of articles that used sensory panel, 26.9% of articles that used e-tongue and 7.1% of articles that used both methods for taste evaluation evaluated a pediatric formulation.

Table 2 shows the articles that used the e-tongue to evaluate the medication taste. Most of the published articles use the e-tongue αAstree. In the pharmaceutical industry, the most commonly used types of e-tongue are: (i) αAstree sensor (Toulouse, France – Patent number: US6290838B1) based on polymer sensors; and (ii) Insent sensor (Atsugi-Chi, Japan – Patent number: US5302262A) based on membrane lipid sensors. The αAstree and Insent sensor are shown in Figure 3. Both sensors measure changes in electronic potential while investigating liquid samples but the underlying sensor technology is different (Eckert et al. 2014). The structure of the membrane is responsible for the sensor answer. The membrane components that are capable of binding to taste substances can cause a variation in the electric potential compared to the human taste receptor (European Medicines Agency 2006). The lipid/polymer membranes are useful transducers of taste information into electrical signals.

The e-tongue αAstree is an analyzer of flavors that performs an overall analysis of a dissolved organic complex or mixture of inorganic compounds, using chemical field effect transistor technology. Flavoring molecules in the sample interact with the proprietary organic coating, which modifies the physical properties of the sensor, resulting in potential variations (Anand et al. 2007). It can be used to evaluate the bitterness of pure active substances in comparison to formulated products. A set of seven sensors evaluate the flavors of medicines. Flavor masking properties are evaluated by comparing the sample signals with those of the placebo formulation and the pure drug using univariate or multivariate statistics (Rachid et al. 2010). The sensors are coated with specific organic membranes with different physicochemical characteristics which interact with ionic and/or neutral compounds present in the sample solutions and distinguish the corresponding flavor. The responses are given in the form of electric potential (Rachid et al. 2010; Preis et al. 2014).

On the other hand, the electronic sensor Insent is used to detect salinity, acidity, sweetness, umami, bitterness and astringency (Chen et al. 2012; Rahman et al. 2012a; Bastiaans et al. 2017). The bitterness of medicines and amino acids has already been evaluated using this system, and the results suggest that the sensor may be useful for this purpose (Hoang Thi et al. 2012; Haraguchi et al. 2016).

The e-tongue systems can be used during customer-oriented product development avoiding consumer complaints and the need for rework in process development, leading to cost savings by the pharmaceutical industry and saving time (Chen et al. 2012).

Eckert et al. (2014) produced granules and mini-tablets for taste improvement. The formulations were investigated utilizing two e-tongues αAstree and Insent, equipped with seven and eight sensors, respectively, which were able to detect differences between all the formulations. In the investigation of Campbell et al. (2012), αAstree detected changes in drug suspension and solution with different flavoring systems for pediatric use.

Table 2. Articles that used the e-tongue to evaluate the taste masking of medications according to searched keywords.

Drug(s)	Dosage form(s)	Objective(s)	Sensor type(s)	Data analysis	Reference
Cetirizine HCl, loperamide HCl, Dextromethorphan HBr	Solution	To compare the water solubility, stability and masking ability of two classes of maltodextrins and cyclodextrins	αAstree and Insent	PCA	(Preis et al. 2014)
Chlorpheniramine tannate complexes	Solution	To evaluate the tannic acid complexes of chlorpheniramine maleate by e-tongue, characterization and dissolution studies	αAstree	PCA and PLS	(Rahman et al. 2012b)
Metamizole sodium and pseudoephedrine sulphate	Solution	To evaluate the results of the release APIs in new formulations using e-tongue	αAstree	PCA	(Wesoły et al. 2018)
485 different drugs	Solution	To investigate the effects of taste masking prediction based upon a database of kinetic parameters of drugs-HP-β-CD generated by surface plasmon resonance imaging	αAstree	PCA	(Guo et al. 2017)
*Isoniazid	Solution	To investigate whether hot melt extrusion could be used to develop taste masked formulations of drug and to evaluate the different taste assessment methods for pediatric use	Insent	PCA	(Keating et al. 2018)
*Brompheniramine maleate	Solution	To evaluate the complexation potential of brompheniramine maleate and tannic acid for sustained release and taste masking effects for pediatric use	αAstree	PCA	(Rahman et al. 2012a)
Diphenhydramine	Solution	To explore the potential applications of the complexation of diphenhydramine (a very bitter tasting drug) and glutamic acid in taste masking and controlled releasing properties	αAstree	PCA	(Agresti et al. 2008)
41 individual substances and mixtures	Solution	To discriminate bitter, sweet and salty substances by using the e-tongue multisensory system	Not informed	PCA and PLS	(Legin et al. 2004)
Chlorpheniramine maleate	Solution	To explore the production of Eudragit® E PO fibers as a taste-masking agent of chlorpheniramine maleate	Insent	PCA	(Abdelhakim et al. 2019)
Ibuprofen	Oral suspension	To show how the composition of a brand product could be deduced and how to rationally develop a taste-masked generic product based on e-tongue measurements	Insent	PCA	(Woertz et al. 2011b)
Ibuprofen and Roxithromycin	Oral suspension	To evaluate the taste of drugs before and after microencapsulation	Not informed	PCA	(Jańczyk et al. 2010)
*Phenobarbital	Microemulsions	Design microemulsions for pediatric administration	αAstree	PCA	(Monteagudo et al. 2014)
Diclofenac	Oral liquid-not specified	To evaluate the ability of the e-tongue to help select a masked flavor formulation	Insent	PCA	(Guhmann et al. 2012)
Ranitidine, Prednisolone, Quinine, Paracetamol, Caffeine, Phenylthiourea and Sucrose octa-acetate	Oral liquid-not specified	To assess the feasibility for taste masking and comparison of taste intensity during formulation development using a multichannel taste sensor system	αAstree	PCA	(Zheng and Keeney 2006)

(*continued*)

Table 2. Continued.

Drug(s)	Dosage form(s)	Objective(s)	Sensor type(s)	Data analysis	Reference
Quinine hydrochloride dihydrate	Oral liquid (not specified)	To explore the impact of the anionic surfactant sodium lauryl sulfate on the sensor response to a model oral liquid formulation containing flavoring agent, preservative, and buffer	Insent	Slopes of sensors responses analysis	(Immohr et al. 2016)
Acetaminophen, Ibuprofen, Tramadol hydrochloride and Sildenafil citrate	Liquid (not specified)	To evaluate the taste masking effect of conventional pharmaceutical sweeteners in different drugs	αAstree	PCA	(Choi et al. 2014)
35 Traditional Chinese medicines	Liquid (not specified)	To evaluate the bitterness of Traditional Chinese medicines	αAstree	PLS	(Lin et al. 2016)
Ranitidine hydrochloride	Liquid (not specified)	To evaluate the ability of the e-tongue to evaluate the taste masking efficiency	Insent	PCA	(Chay et al. 2018)
*Oseltamivir phosphate	Liquid (not specified)	To evaluate a pediatric-friendly formulation of a bitter tasting drug	αAstree	PCA and PLS	(Siddiqui et al. 2013)
*Ambroxol hydrochloride, cetirizin dihydrochloride and Sodium picosulfate	Liquid (not specified)	To prove suitability of e-tongue combined with multivariate data analysis for taste-masking evaluation in pediatrics	Insent	PCA and PLS	(Immohr et al. 2017b)
*Cefpodoxime proxetil	Lyotropic liquid crystalline-not specified	To develop a colloid-stable cefpodoxime proxetil-loaded lyotropic liquid crystalline nanoparticle system with taste-masking property	Insent	PCA	(Fan et al. 2020)
Tacrolimus phospholipid	Nanodispersions	To explore the potential of quality by design in understanding the formulation of phospholipid-stabilized tacrolimus nanodispersions by microfluidization	αAstree	Discriminant functional analysis (DFA)	(Zidan 2017)
Lidocaine HCl	Injectable	Masking the bitter taste of dental lidocaine HCl injection using hydroxypropyl-β-cyclodextrin and sodium saccharin	αAstree	PCA	(Wei et al. 2015)
Quinine hydrochloride dihydrate and Ibuprofen sodium dehydrate	Tablet	To study potentiometric e-tongue system as a possible candidate for in-line dissolution profiling	Not informed	PLS	(Khaydukova et al. 2017)
Sanhuang (Traditional Chinese herbal medicine)	Tablet	To develop the powder coating technology to mask the bitter taste of Sanhuang tablets	αAstree	PCA	(Feng et al. 2019)
Valsartan	Mini-tablet	To investigate the release of valsartan from mini-tablets coated with Eudragit® using E-tongue systems	αAstree and Insent	PCA	(Wesoły et al. 2016b)
*Sodium benzoate	Mini-tablet	To develop a child appropriate, taste masked dosage form, utilizing lipids in melt granulation process and tableting	αAstree and Insent	PCA	(Eckert et al. 2014)
*Ketoprofen	Mini-tablet	To develop swallowable dosage forms, primarily to be used by pediatric and geriatric patients	αAstree	PCA	(Alshetaili et al. 2016)
Diclofenac	Orodispersible tablet	Diclofenac taste-masked orodispersible tablets development with fast release characteristics	Insent	PCA	(Guhmann et al. 2015)
Metformin hydrochloride	Orodispersible tablet	To evaluate the viability of the production of orodispersible tablets	Insent	PCA	(Petrovick et al. 2018)

(*continued*)

Table 2. Continued.

Drug(s)	Dosage form(s)	Objective(s)	Sensor type(s)	Data analysis	Reference
Cyclosporine	Orodispersible tablet	To investigate the effects of formulation parameters on the disintegration, water absorption and dissolution characteristics of cyclosporine	αAstree	PCA	(Zidan et al. 2015)
Lamotrigine-cyclamate salt	Orally disintegrating tablet	Design and evaluation of orally disintegrating tablet of novel lamotrigine-cyclamate salt. The tablets were characterized, and taste masking evaluation was performed	αAstree	PCA and PLS	(Rahman et al. 2013)
Sildenafil citrate	Orally disintegrating tablet	To produce taste masked formulations of sildenafil citrate using hot-melt extrusion technology	αAstree	PCA	(Morott et al. 2015)
*Dimenhydrinate	Orodispersible film	To develop a taste-masked orodispersible film for the pediatric population	αAstree and Insent	PCA	(Preis et al. 2012)
Betahistine hydrochloride	Orodispersible film	To develop a slow release orodispersible film	Insent	Cluster analysis	(Shang et al. 2018)
Ritonavir	Oral solid granules	Development of a novel lipid-based nanotechnology to formulate poorly water-soluble drugs in oral solid granules to improve stability, palatability and bioavailability	αAstree	PCA	(Guo et al. 2016)
Praziquantel	Polymer microcapsules and microspheres	To develop polymeric microparticulate systems to encapsulate a drug to be used in a pharmaceutical formulation for oral use	Not informed	PCA	(Machado et al. 2018)
Roxithromycin	Solid oral (not specified)	To obtain microencapsulated roxithromycin with highly efficient taste masking effect of this bitter drug	Not informed	PCA	(Sollohub et al. 2011)
Itraconazole	Solid oral-not specified	To obtain an optimized itraconazole solid oral formulation in terms of palatability and dissolution rate	αAstree	PCA	(Malaquias et al. 2018a)
*Propranolol hydrochloride	Solid (matrix microbeads)-not specified	To develop formulations with the taste masking agent Eudragit® E PO using the prilling technique	Not informed	PCA	(Lopalco et al. 2020)
Fluconazole	Solid inclusion complexes-not specified	To mask fluconazole taste and improve its rheological properties by cyclodextrin complexation	αAstree	PCA	(Malaquias et al. 2018b)
*Acetaminophen	Powder	To produce and characterize taste-masked powders for pediatric use	αAstree	PCA	(Hoang Thi et al. 2012)
Quercetin	Powder	To microencapsulate drug powders using hot-melt extrusion aiming at taste masking	Insent	Bitterness sensory output	(Khor et al. 2017)
*Quinine sulphate	Pellets	To develop a taste-masked quinine sulphate dosage form as a flexible pediatric formulation tool	αAstree	PLS	(Kayumba et al. 2007)
Metformin HCl	Pellets	To investigate the taste-masking properties of metformin HCl lipid based micropellets obtained by solid lipid cold extrusion followed by the spheronization method	Insent	PCA	(Petrovick et al. 2016)
*Mefloquine	Orodispersible tablet and oral liquid	To develop a stable, rapidly dissolvable and palatable pediatric formulation for a poorly soluble and highly bitter drug using liposomes	αAstree	PCA	(Tang et al. 2017)

(*continued*)

Table 2. Continued.

Drug(s)	Dosage form(s)	Objective(s)	Sensor type(s)	Data analysis	Reference
A hydrophilic and a hydrophobic model drug (not specified)	Pediatric suspension, solution and powder for reconstitution	To develop different pediatric formulations and to evaluate the taste	αAstree	PCA	(Campbell et al. 2012)
Cetirizine dihydrochloride	Powder, microsphere and lyophilizate	To evaluate the influence of various production process on taste masking efficiency	Not informed	PCA	(Wesoly et al. 2016a)
Quinine hydrochloride	Not specified (intermediate product)	To evaluate the feasibility of the use of hot melt extrusion for the complexation of the drug in generating a product with masked flavor	Insent	PCA	(Tan et al. 2018)
Metamizole sodium and pseudoephedrine sulphate	Some dosage forms (not specified)	To investigate the influence of excipient mixture on the electronic tongue analysis of active pharmaceutical ingredients	αAstree	PCA	(Wesoły et al. 2017a)

*Only pediatric dosage form use.

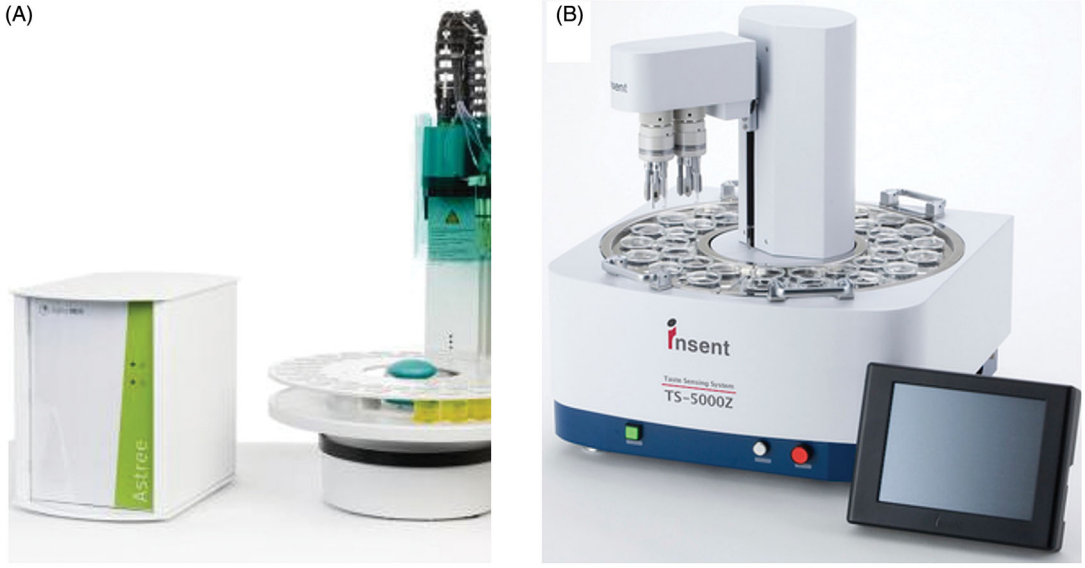


Figure 3. Commercial electronic tongues: A: α-Astree and B: Insent.

The most used methods for data analysis in the e-tongue are Principal Component Analysis (PCA) and the Partial Least Squares method (PLS). The PCA method provides a two-dimensional or three-dimensional chart in which a formulation can be considered a pleasant common taste if the distance from the placebo formulation is small. However, unsuccessful flavor masking is expected if the distance to the pure active pharmaceutical ingredient is small. These distances between the formulations and the pure drug can be calculated as Euclidean distances (Rachid et al. 2010). The principle of operation, components, types of sensors and data analysis for these sensors have been extensively described before (Duong et al. 2002; Jain et al. 2010; Rachid et al. 2010; Tang et al. 2017). The PLS is a multivariable statistical method similar to PCA, which estimates a linear regression model projecting the observable and predicted variables into a new space. Using this method, the flavor is described in a measurable way (Maniruzzaman et al. 2014).

A consequence of using the PCA technique is that the results obtained are based on theoretical assumptions that must be true. The technique is an approach to reduce dimensionality, which, in this way, represents, but not in full, the variability of the studied phenomenon. This is dependent on the collection and quality of the data. In this way, a loss of information is possible with the use of the technique.

As seen in Table 1, the most cited dosage form was solution. This is one of the most commonly used dosage forms for children, especially when the administration is via the oral route, which is less invasive and more practical when compared to other routes (European Medicines Agency 2006).

Preis et al. (2014) verified that maltodextrins, analyzed by e-tongue, could be a safe taste mask excipient, e.g. in syrup development for pediatric use with good taste. Rahman et al. (2012b) showed that the tannate complex of chlorpheniramine maleate masked the bitter taste of the drug, which would make it more patient friendly. In these studies, the developed formulations are not exclusively for pediatric use. However, the term 'pediatric' is cited in the introduction, results, and conclusion, respectively, showing that pleasant taste formulations may be used in pediatrics because it improves the adhesion of these patients.

On the other hand, Hoang Thi et al. (2012) developed and evaluated taste-masked acetaminophen (analgesic and antipyretic) for the pediatric public. With a similar purpose, Tang et al. (2017)

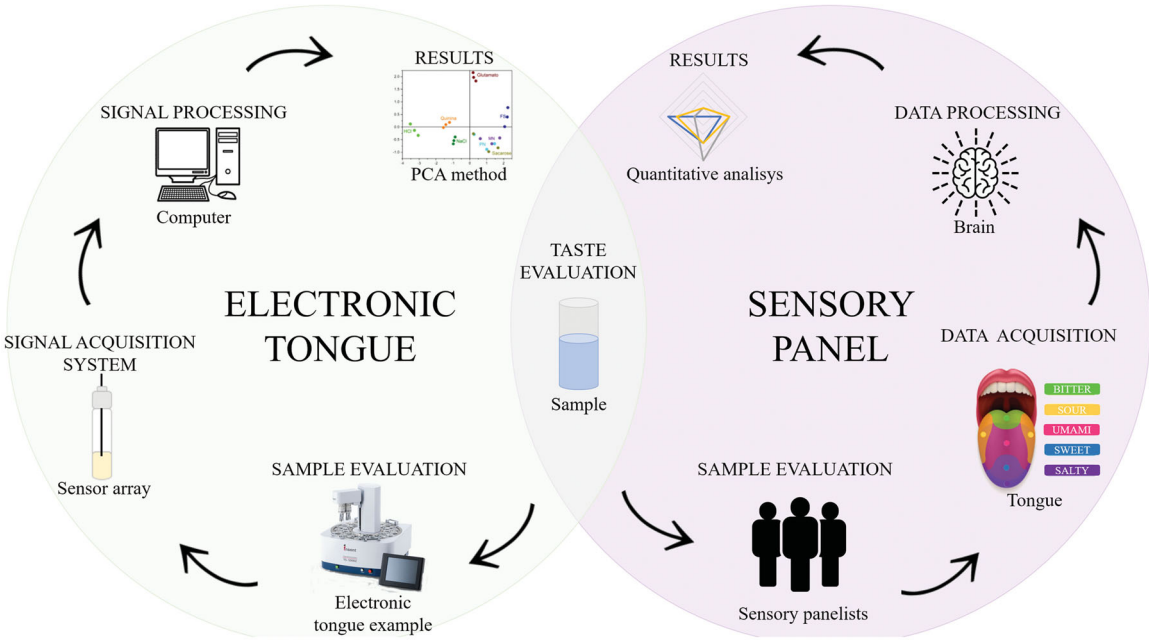


Figure 4. Scheme showing the data processing steps by the two main methods of taste evaluation.

developed a rapidly dissolved oral pediatric formulation for mefloquine (antimalarial) delivery using liposomes. Both studies have succeeded in the taste masking of their pediatric formulations using different approaches, evaluated by e-tongue.

Combined methods

Researchers would consider that taste assessment methods are complementary to each other and, it is appropriate, when possible, to carry out the two assessment methods to bring more reliable data. In this sense, the study by Immohr et al. (2017b) shows in practice the importance of the complementarity of the electronic tongue and/or human sensory panel methods for pediatric formulation development.

Figure 4 shows in a simplified way the analogous data processing steps exemplified by the two main methods of flavor evaluation. Each technique has advantages and disadvantages. Depending on the sample to be analyzed, one technique can stand out from the other. For example, if the analysis is of a highly toxic substance, the electronic tongue technique becomes more applicable, and if the sample is a tablet or mini-tablet, the sensory panel technique stands out, since there is no need to solubilize the sample before analysis.

The simplicity, the speed, the low-cost and the possibility of smaller amounts of samples are some advantages regarding the analysis of flavor with electronic taste detection devices (Woertz et al. 2011a; Thompson et al. 2015). This detection system can safely reduce the dependence on human taste assessment panels. Besides that, the electronic tongue has less bias influence (for example, loss of interest or possible distractions); more precision in descriptive data; easier interpretation of results; no fatigue like the human tongue; absence of risks and no human error if the result is well interpreted (Sipos et al. 2013; Newman et al. 2014; Thompson et al. 2015).

Although electronic tongues offer interesting advantages, there are several restrictions that can limit the overall clinical utility in determining the final acceptability of a particular pharmaceutical formulation (Thompson et al. 2015). Electronic devices with taste sensors do not contain biological receptors and, therefore, may not be accurate in adequately and safely reporting the degree of bitterness. Likewise, sensory devices use normative values to identify sensory extremes and isolates the characteristics by the sensors. In this way, the electronic devices may not truly identify perceptions and preferences associated with human senses. Finally, electronic detection devices are unable to provide qualitative feedback on physical characteristics related to palatability and ease of ingestion, such as dry mouth, indigestion, or nausea (Haraguchi et al. 2016).

The human sensory panel has disadvantages such as the subjectivity of the method, the difficulty to explain the process and a greater difficulty in authorization, due to the approval of an Ethics Committee. In addition, there is a greater difficulty depending on the age of the volunteers, sample toxicity, adverse reactions, pre-existing diseases and execution of the process (Thompson et al. 2015).

The human sensory panel can be able to meet these requirements that are lacking in the use of an electronic device, being considered a more realistic method of the degree of human flavor, with its inherent variability in this process. However, disadvantages such as the subjectivity of the method, the difficulty to explain the process and a greater difficulty in authorization, due to the approval of an Ethics Committee, should be considered. In addition, there is a greater difficulty depending on the age of the volunteers, sample toxicity, adverse reactions, pre-existing diseases and execution of the process (Thompson et al. 2015).

Therefore, combining the techniques and evaluating the results is considered a more efficient strategy. The ultimate goal of more recent research is to get the ideal solution by reducing or inhibiting bitterness, and this can perhaps be achieved by discovering the universal inhibitor of all bitter tasting substances that do not affect other flavors like sweetness. However, to date, a substance that acts as a universal inhibitor of a bitter taste has not yet been discovered. To try to confirm the taste of the drug and some excipients of pharmaceutical forms, many studies are using two or more techniques concomitantly, generally using an *in vivo* method, such as the sensory and an *in vitro* method such as the e-tongue.

In the study by (Wesoły et al. 2017b) in which cetirizine microspheres were prepared with the use of two coating agents with different ratios of drug and polymer for flavor masking, high correlation of the results of the e-tongue and the human panel was found. This study demonstrated similar results regarding the use of the sensory panel and the use of the e-tongue. A good correlation between e-tongue and human panel was also obtained by (Abu-Khalaf et al. 2018), when evaluating the taste of clarithromycin oral pharmaceutical suspensions using αAstree and comparing to an *in vivo* taste assessment conducted on paediatric patients and volunteering community pharmacists. The capability of the e-tongue in the taste assessment of oral suspensions was confirmed, and the obtained results were in agreement between the patient and pharmacist.

Although the results obtained from an *in vitro* method have corresponded to human panel, and often, with a high correlation, this method does not actually replace the *in vivo* method. Sadrieh et al. (2005) showed that the e-tongue was able to differentiate the flavors but classified the masking agents in a different order than human volunteers. The authors justify this difference based on the diverse sources for obtaining the elements and that the e-tongue would serve as a preliminary method to eliminate the most inappropriate samples, serving as a screening to expose less the panelists in the sensory test. A similar conclusion was observed by Bastiaans et al. (2017). In this study, the palatability of a new paediatric valaciclovir formulation was tested in children and their parents and the results were compared with an electronic tongue assessment. The authors concluded that the number of children needed for a comparative taste assessment can be lowered by application of an electronic tongue measurement as screening during the development of a new paediatric formulation.

These two tests are complementary and the combination of both confirm that improvement in the taste of a pharmaceutical formulation is more efficient, according to some related studies.

Table 3 shows the articles that used both methods to evaluate the taste of different formulations. It can be seen that the most analyzed drug regarding taste using both assessment techniques is cetirizine dihydrochloride, a drug used in the treatment of allergic disorders, indicating that such a substance has an unpleasant taste and that it is frequently studied with the aim of improving its taste. These studies have resulted in different dosage forms of this drug.

Kim et al. (2013) and Li et al. (2007) used Amberlite IRP 64 as a polymer for flavor masking of different drugs. Both studies used αAstree as an e-tongue sensor, besides using human panel, and showed satisfactory results in taste masking. The difference between the two studies was that Kim et al. (2013) focused on the drug donepezil hydrochloride using 12 volunteers and Li et al. (2007) analyzed some drugs regarding taste masking using 5 volunteers. Only one article in Table 3 used a child sensory panel, applying a hedonic scale along an e-tongue evaluation (Bastiaans et al. 2017). This is the first study in which results from palatability testing in children and adults as well as results from an e-tongue assessment are simultaneously evaluated.

Human panel tests involve sensitive analyses, built on statistical designs that minimize bias and response variations among human volunteers. These tests generate information about the individual's attitudes and expectations and can generate important information to guide the development of new formulations (Anand et al. 2008). Thus, sensory analysis is an essential part of the development and quality control of oral formulations that are adapted to the need of special populations, such as that of the pediatric (Anand et al. 2008; Kobayashi et al. 2010; Joshi and Petereit 2013).

Some studies found in the literature use the evaluation of acceptability in children through their caregivers. Those in charge of administering the drugs are usually asked to take a form home or to keep a journal, where they record the child's acceptance at each administered dose of the formulation being analyzed using a scale. This procedure has been used in children aged 1–7 years old, who used a three-point scale ('very good', 'good' and 'bad') to evaluate the taste of an antimicrobial in different flavors and, consequently, their acceptability in this age group (Jahnsen and Thorn 1987). Martinez et al. (2006) and Lava et al. (2011) determined the acceptance through a test applied to the mother, immediately after the administration of different formulations to newborns, in which she evaluated the child's reaction in a hedonic scale, performing an indirect test. In Canada, children usually receive pediatric formulations, while in developing countries, tablets designed for adults are often broken or crushed for children, compromising the effectiveness of the treatment (Haraguchi et al. 2014).

Palatability testing in children is complex and difficult, coupled with many practical and technical challenges, including recruitment of subjects (considering age range, ability to communicate, complexities of palatability and reliability with responses), and of ethical questions about children's participation in drug-related studies (Zajicek et al. 2013). In the case of children under 4 years of age who are unable to respond to acceptance tests alone, assessment by the primary caregiver is a good source of information about the palatability of formulations.

The oral route is the most preferred for better patient compliance and easy administration especially when it comes to children. However, this administration needs to be pleasurable for treatment adherence. Due to the stricter ethical issues regarding children, the flavor analysis is often performed only by adults, even if the drug/medication is designed for children (Matsui 2007).

There are few studies that focus on the evaluation of taste masking for pediatric use in children, though one example is the investigation performed by Cheung et al. (2018). This study used *in vitro*, *in vivo* and clinical data to evaluate the palatability of a novel midazolam chocolate tablet. They used a visual analogue scale score of patients aged 4–16 years prescribed with midazolam pre-surgery.

The quality of taste studies depends on the training and ability of the assessors and how the samples are presented. The ability to communicate is also important, especially if descriptive analysis is required. Usually, when adults are used in the human panel, they are preselected for their ability to perform a sensory test, being previously trained for such analysis. However, with children it is more complicated because children are not ideal assessors. Their attention span is limited; and taste fatigue or loss of concentration are frequent issues (Joshi and Petereit 2013). In addition, there are ethical issues involving children, which are always more complicated when compared to adults. Therefore, it is always easier to use adults, in the case of a human sensory panel.

Another comparative study by Sadrieh et al. (2005) showed that the e-tongue was able to differentiate the flavors, but classified the masking agents in a different order than human volunteers. The authors justify this difference based on the diverse sources for obtaining the elements and that the e-tongue would serve as a preliminary method to eliminate the most inappropriate samples, serving as a screening to expose less the panelists in the sensory test.

Table 3. Articles that used both human panel (sensory analysis) and e-tongue to evaluate the taste of medications according to searched keywords.

Drug(s)	Dosage form(s)	Objective	Human panel	E-tongue/data analysis	Reference
Propiverine hydrochloride	Orally disintegrating tablets	To demonstrate the usefulness and wide applicability of a taste sensor and a new disintegration testing apparatus in the development and evaluation of orally disintegrating tablets comparing to a human panel of bitterness	6 well-trained volunteers (ages not declared). Numerical scale (0–5)	Taste-responding system SA402B (Taste sensor) of intelligent sensor technology, Ltd., Japan/PCA	(Harada et al. 2010)
Berberine hydrochloride	Solution	To predict the bitterness intensity of a drug using an e-tongue and volunteers	20 healthy volunteers were selected as the subjects (9 male and 11 female), (ages not declared). Numerical scale (0–5)	αAstree/PLS	(Liu et al. 2014)
H1–antihistamines	Solution	To predict a drug's bitterness and estimate the taste masking efficiency using an e-tongue with artificial sweeteners and volunteers	11 well-trained volunteers (ages not declared). Numerical scale (0–4)	αAstree/PCA and PLS	(Ito et al. 2013)
Acetaminophen dextromethorphan, phenylephrine, diphenhydramine and oxybutynin chloride	Solution	To compare drug formulations using human taste panel and e-tongue data. Multiple formulations of the same bitter drug were evaluated and compared to commercial products	4–6 trained pharmaceutical sensory panelists (ages not declared). Scale: 1-no bitterness, 2-very slight bitterness, 3-slight bitterness, 4-slight to moderate bitterness, 5-moderate bitterness, 6-moderate to strong bitterness, and 7-strong bitterness	αAstree/PCA and PLS	(Lorenz et al. 2009)
Doxycycline, ciprofloxacin hydrochloride and potassium iodide	Solution	To evaluate the ability of common household food and drink products to mask the bitter taste of drugs	30 healthy adult volunteers participated in each study. 90% of volunteers were in the age group from 20 to 29. Numerical scale (0–5)	αAstree/PCA	(Sadrieh et al. 2005)
Cetirizine dihydrochloride	Solution	To verify the correlation of the e-tongue results and human panel scores from microspheres prepared with the use of two coating agents	6 healthy volunteers (ages not declared). Scale: 0-no bitter; 1-slight bitter; 2-moderately bitter and 3-very bitter	αAstree and Insent/PCA	(Wesoły et al. 2017b)
Rupatadine fumarate	Solution	To develop taste-masked microparticles of rupatadine fumarate	6 healthy volunteers (ages not declared). Scale: 0-not bitter, 1-slightly bitter, 2-moderately bitter, 3-very bitter	A self-constructed electronic taste sensing system (electronic tongue) was used in this study/PCA	(Wasilewska et al. 2019)
Clarithromycin, erythromycin, cefdinir, vancomycin hydrochloride, doxycycline hydrochloride, tetracycline hydrochloride, oxytetracycline hydrochloride, minocycline hydrochloride, bacampicillin hydrochloride	Solution and suspension	To evaluate the bitterness of commercial antibiotics in human gustatory sensation tests with volunteers and to compare the results with e-tongue	9 volunteers (ages not declared). The bitterness scores were defined as − 1, 0, 1, 2, 3 and 4	Taste-sensing system SA402B of intelligent sensor technology Co. Ltd (Atsugi, Japan/PCA	(Uchida et al. 2003)
*Clarithromycin	Suspension	To evaluate the taste of clarithromycin oral pharmaceutical suspensions	The first part of this study included pediatric patients aged 3–12 years and for the second part of this study, 150 community pharmacists were invited to evaluate the taste. Hedonic Scale	αAstree/PCA	(Abu-Khalaf et al. 2018)

(*continued*)

Table 3. Continued.

Drug(s)	Dosage form(s)	Objective	Human panel	E-tongue/data analysis	Reference
Efavirenz	Liquid (not specified)	To develop and characterize a stable, taste masked self-emulsifying drug delivery system	10 healthy human volunteers (age 22–27 years). Scale: 0-no bitterness, 1-tasteless, 2-slightly bitter, 3-moderately bitter, and 4-extremely bitter	e-Tongue (fabricated in-house in Government College of Engineering, Pune, India)/PCA	(Rao et al. 2019)
Cetirizine hydrochloride and verapamil hydrochloride	Extrudates (not specified)	To investigate the potential of hot-melt extrusion for masking bitter taste using different polymers	67 healthy human volunteers (age 18–25). Scale from 1 to 5, where 1, 2, 3, 4 and 5 indicate none, threshold, moderate, bitter and strong bitterness	αAstree/PCA and PLS	(Maniruzzaman et al. 2014)
Gankeshuangqing (Traditional Chinese Medicine)	Dispersible tablet	To use particle coating technology to mask bitter taste	20 well-trained and healthy volunteers (half men and half women, age 22–28). Numerical scale (0 − 4)	αAstree/PCA	(Han et al. 2017)
Caffeine citrate	Orally disintegrating tablet	To use hot-melt extrusion technology aiming at masking unpleasant bitter taste	9 human subjects (age 18–42). Modified hedonic scale where (0) corresponds to no taste, (1) threshold, (2) slightly bitter, (3) moderately bitter, (4) bitter and (5) strongly bitter	αAstree/PCA	(Pimparade et al. 2015)
Ebastine	Orally disintegrating tablet	To analyze the palatability of ebastine orally disintegrating tablets	6 healthy female subjects (29 ± 7 years old). Numeric scale (0–4)	αAstree and Insent PCA	(Haraguchi et al. 2014)
Donepezil hydrochloride	Orally disintegrating tablet	To prepare an orally disintegrating tablet containing donepezil HCl complexed with ion exchange resin, AmberliteIRP-64 for rapid disintegration and bitter taste masking	12 human healthy volunteers (age 24–31). Numerical scale (0–5)	αAstree	(Kim et al. 2013)
Famotidine and amlodipine	Orally disintegrating tablet	To evaluate orally disintegrating tablets by an e-tongue and a human panel	31 healthy adults (16 men and 15 women; mean age standard 22.1 years). Visual Analogue Scale (100 mm)	αAstree/PCA and PLS	(Nakamura et al. 2015)
A weakly basic model drug (not specified)	Orally disintegrating tablet	To analyze taste masking properties using Amberlite IRP 64 and Carbopol	5 healthy volunteers (ages not declared), Scale: Score 0 indicates no bitterness sensation, + to +++ indicate the increase in bitterness intensity	αAstree/PCA	(Li et al. 2007)
Cetirizine dihydrochloride	Orally disintegrating tablets and lyophilizates	To prepare orally disintegrating tablets and oral lyophilizates containing taste masked microparticles	6 healthy volunteers (ages not declared). Scale: 0-pleasant/not rough; 1-slightly pleasant/slightly rough; and 2-unpleasant/rough	Sensor array of the electronic tongue was composed of 16 ion-selective electrodes (ISEs) of various cross-selectivity/ PCA	(Amelian et al. 2017b)
Cetirizine dihydrochloride	Orodispersible dosage forms	To formulate and characterize microparticles obtained by the spray drying method with cetirizine and methacrylate as a barrier coating	6 selected healthy volunteers (ages 19–21). Scale: 0-not bitter; 1-slightly bitter;2-moderately bitter and 3-very bitter	An electronic tongue based on 16 potentiometric standard ion-selective electrodes (ISEs): 12 cation-selective and 4 anion-selective/PCA	(Amelian et al. 2017a)
Berberine hydrochloride	Powder	To apply a powder surface modification technology to mask the bitterness of the drug and improve patient compliance	20 well-trained and healthy volunteers (ten males and ten females, age 18–28). Volunteers were trained with different concentrations of standard solutions (0, 0.027, 0.134, 0.269, and 1.345 mmol L^{-1}) to accustom them to evaluation scales and intensity of bitter taste of the standard solutions	αAstree/PCA	(Jiang et al. 2018)

(*continued*)

Table 3. Continued.

Drug(s)	Dosage form(s)	Objective	Human panel	E-tongue/data analysis	Reference
Epinephrine	Sublingual tablet	To assess the potential of the e-tongue to determine the degree of epinephrine bitterness and to evaluate the taste-masking effect of sweetening and flavoring non-medicinal ingredients	The human panel was used to examine the correlation between *in vivo* measurements and the e-tongue measurements using a Standard Model, as Provided by Alpha M.O.S.	αAstree/PCA and PLS	(Rachid et al. 2010)
*Valaciclovir	Crushed and suspended tablet	To investigate the palatability of a new pediatric formulation for children and their parents and compare to the reference	($n = 20$) All subjects tasted 4 mL (children 4–8 years) or 8 mL (children 8–12 years and the parent). Use of hedonic scale	Insent/PCA	(Bastiaans et al. 2017)
Propranolol hydrochloride	Solid (not specified)	To investigate the taste masking effect of hot melt extruded formulations by e-tongue and a sensory panel evaluation	6 healthy human volunteers were selected (male = 3, female = 3) , (ages 18–25). Numerical scale (0–5)	αAstree and Insent/PCA and PLS	(Maniruzzaman and Douroumis 2015)
Diclofenac	Fast-dissolving film	To evaluate the suppression of the bitter taste of a maltodextrin fast-dissolving film loaded with sodium diclofenac by adding taste-masking agent	10 healthy volunteers (ages not declared). Numerical scale (0–4)	Taste-Sensing system SA 402B (Intelligent sensor technology Co. Ltd, Japan)/PCA	(Cilurzo et al. 2011)
Nicotine	Fast dissolving film	To develop a fast-dissolving film made of low dextrose equivalent maltodextrins containing nicotine hydrogen tartrate salt	10 healthy volunteers (ages not declared). Numerical scale (0–4)	Taste-Sensing System SA 402B (Intelligent Sensor Technology Co. Ltd, Japan)/PCA	(Cilurzo et al. 2010)
Cetirizine hydrochloride	Solid oral dosage forms (not specified)	To investigate whether cyclodextrins or ion exchange resins could be used to taste-mask cetirizine when formulated in a freeze-dried oral formulation	30 volunteers taste panel (aged 18–63). Hedonic and Just About Right (JAR) scales	Insent/PCA	(Preis et al. 2015)
Sildenafil citrate	Microcapsule	To prepare a taste-masking system to mask bitterness of drug using microcapsules	6 well-trained and healthy volunteers (age 24–28, 4 males and 2 females). Scale of 0–4 (0-no taste, 1-threshold, 2-slightly bitter, 3-bitter, and 4- remarkably bitter)	αAstree/PCA	(Yi et al. 2014)
Nizatidine	Ion-Exchange Resins (not specified)	To mask the bitter taste and formulate orodispersible tablets	6 healthy adult volunteers (3 males and 3 females, with a mean age of 25.7 years). Scale: 0-pleasant/not bitter; 1-slightly pleasant/slightly bitter; and 2-unpleasant/bitter	αAstree/PCA	(Panraksa et al. 2019)

*Only pediatric dosage form use.

Conclusion

A systematic review of the literature was conducted to evaluate articles that used e-tongue and/or sensory analysis in the assessment of medicines taste, in a way that a total of 88 articles were found. The number of publications in recent years regarding this subject has been growing. The e-tongue (57.5%) has been more frequently described than the sensory panel (10.3%), while some articles (32.2%) used both techniques. 74.7% of the articles mentioned the terms 'pediatric', 'paediatric' and/or 'child' in some part of the text.

The use of a sensory panel is much more complex than the e-tongue analysis, especially when considering the participation of children in these studies, due to the ethical and toxicity issues involved. Achieving a statistically significant number of volunteers often also becomes an obstacle to reaching a reliable outcome in the research. Despite the several advantages of the e-tongue, it is not capable of completely substituting the *in vivo* test. In this sense, it is proposed the use of the e-tongue as a fast way to select the most promising sample(s) and, after that, the sensory panel should be applied in order to confirm the taste masking.

Perspectives

With the advancement of flavor modification techniques in the areas of scientific research as well as food and pharmaceutical industry, the importance of flavor evaluation is continually increasing. The taste evaluation tests are constantly evolving, with much more emphasis given to the ethical rules of the participants. Regulatory guidelines for studies of taste evaluation specifically in pediatric patients need more attention. Despite emerging *in vitro* approaches, taste assessment tests are and will remain a standard approach. In this sense, one technique becomes complementary to the other.

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